SN Computer Science (2024) 5:1028
https://doi.org/10.1007/s42979-024-03369-0

ORIGINAL RESEARCH

Intrusion Detection: A Comparison Study of Machine Learning Models Using Unbalanced Dataset

Sunday Adeola Ajagbe[1,2] · **Joseph Bamidele Awotunde**[3] · **Hector Florez**[4]

Received: 4 September 2024 / Accepted: 20 September 2024
© The Author(s) 2024

Abstract

The worldwide process of converting most activities of both corporate and non-corporate entities into digital formats is now firmly established. Machine learning models are necessary to serve as a tool for preventing illegal intrusion onto different networks. The machine learning (ML) model's strengths and drawbacks pertain to intrusion detection (IDS) tasks. This study used an experimental methodology to assess the efficacy of various ML models, including linear SVC, LR, random forest (RF), decision tree (DT), and XGBoost, in detecting intrusion on the UNSW NB15 datasets. The objective is to compare the strengths and shortcomings of these models. Data exploration, Feature engineering, selection and a test set of 15%, a validation set of 15%, and a training set of 70% respectively were used for data splitting. Performance evaluation was carried out using accuracy, recall, precision F1-score and confusion matrix plotted. The outcome of the experiment shows a percentage of 92.71% (1, normal) and 7.29% (0, attack) for normal traffic and attack traffic respectively. Performance evaluation results showed that RF and XGBoost outperformed the other ML models. Hence, ML models can effectively be used to detect system attacks. We intend to expand this research in the future and use the paradigm in a real-world setting with further conclusions and justifications.

Keywords Comparative analysis · Contemporary machine learning · Deep learning · Detection experiment · Intrusion detection systems (IDSs)

Background

A branch of internet of things (IoT) called industrial IoT (IIoT) is centred on industrial assets and manufacturing process automation. IIoT is closing the gap between information technologies and operational technologies by integrating control and information systems with physical and business operations [1, 2]. In the IoT, intrusion detection systems (IDSs), are made to identify and guard against malicious activity or unauthorized access to a network, system, or application. They are useful for identifying and stopping any harmful IoT activity. Additionally, they can assist in locating any breaches and attacks, enabling a company to take the necessary precautions before more harm is done. Large volumes of data can be analyzed by machine learning ML model to find trends that might point to attacks [3, 4].

The security of data is at risk whenever the IDS used to identify network breaches are not handled securely or when advanced techniques like machine learning (ML) attempts to enhance these systems are compromised. Various classification techniques are employed to identify different types of attacks, hence enhancing the efficiency of IDS. Choosing the optimal solution for constructing an IDS is a challenging endeavor [5]. The most effective approach is to evaluate the various classification algorithms' performance. However, as a measure of classification ability, the confusion matrix has drawn the attention of the majority of studies. As a result,

✉ Sunday Adeola Ajagbe
saajagbe@pgschool.lautech.edu.ng

Joseph Bamidele Awotunde
awotunde.jb@unilorin.edu.ng

Hector Florez
haflorezf@udistrital.edu.co

1 Department of Computer Engineering, Abiola Ajimobi Technical University, Ibadan 200255, Nigeria

2 Department of Computer Science, University of Zululand, Kwadlangezwa 3886, South Africa

3 Department of Computer Science, University of Ilorin, Kwara State, Ilorin 240003, Nigeria

4 Universidad Distrital Francisco Jose de Caldas, Bogotá, Colombia

Published online: 06 November 2024

this matrix is frequently used in publications to provide a thorough comparison with the data preprocessing, dataset, feature selection and classification methods, and performance assessment [6]. The main goal of this work is to compare the utilization of modern ML models for constructing and enhancing IDSs. The comparison will be based on metrics such as confusion matrix, accuracy, recall, precision, and F1-score. The UNSW-NB15 Dataset will be employed for this analysis. Therefore, the following is a summary of the study's primary contributions:

(i) Compare the contemporary ML models viz-a-viz linear SVC, LR, random forest (RF), decision tree (DT) and XGBoost on IDS using unbalance dataset (UNSW NB15).
(ii) To performance evaluation of the ML models based on metrics such as detection accuracy, recall, precision, F1-score, area under the curve (AUC), receiver operating characteristic (ROC), and confusion matrix.

The remaining parts of this study are arranged as follows: The IDSs employing ML-based model comparison and DL-based are covered in the sect. "Review of Related Works"of the Review of related works. The Methodology, which portrays the design and implementation setup for the ML-based IDSs comparative research is presented in sect. "Methodology". The Experimental results and discussion display the implementation and results in sect. "Result and Discussion". The Conclusion that wraps together the study and suggests future work is contained in sect. "Conclusion"

Review of Related Works

A review of related works in comparative analysis of ML is presented in this section. In Ref. [7], the authors attempted construction of relevant, knowledgeable, and intelligent IDS, numerous ML approaches were created, including Naive Bayse (NB), LR, DT, SVM, k-NN, and meta-heuristic techniques. The IDS has not improved enough recently to keep up with all new threats and safeguard the devices. Consequently, in order to find potentially reliable approaches, it is imperative to research ML techniques in the security sector.

In order to identify the multi-class intrusion attacks in the IoT, Saran and Kesswani [8] in their study introduced an IDS based on the multiple ML classifier methods on the message queuing telemetry transport-IoT-IDS dataset. A comparison of the results of the result analysis has also been done in this research work to assess the effectiveness of all ML classifier techniques that have been employed. The tested IDS's overall accuracy was 96.76%, 97.38%, 99.98%, 99.98%, and 97.58% when employing the classifiers k-NN, SVM, NB, RF, DT, and stochastic gradient descent (SGD), in that order. All of the tested ML classifiers' recall, precision, and F1-score are listed in the results and analysis section [8]. A set of balanced datasets was used in the investigation conducted by [8] which makes it defer from our study even though both studies have similar focus.

To identify network- and application-based threats, Nizamudeen [9] established an intelligent intrusion detection framework. The three stages of the suggested framework are feature selection, classification, and data pre-processing. For the feature selection stage, the opposition-based learning-rat inspired optimizer (OBL-RIO) was intended. Rats' evolutionary tendency selects the important characteristics. The stability of the OBL-RIO characteristics is guaranteed by the fittest value. As the binary class classifier, a 2D array-based convolutional neural network was suggested. To operate on complicated layers, a 2D-array model maintains the input features. It recognizes both typical and unusual traffic. The Netflow-based datasets were used to train and evaluate the suggested framework. 95.2% accuracy, 2.5% false positive rate, and 97.24% detection rate are produced using the suggested framework [9]. Reference [9] study conducted a balanced dataset.

Rbah et al. [10] used a variety of ML classifiers, such as K-nearest neighbours (KNN), SVM, RF, DT, Gaussian NB, XGB, LR, artificial neural networks, and multilayer perceptron (MLP), to detect different types of network attacks on IoT devices and to find the best model that performs well in identifying suspicious patterns of IoT network activity. These models have been evaluated by the IoT Botnet dataset using performance metrics like prediction time and accuracy. As a consequence, when the MLP algorithm was used for multiclass traffic recognition, the examined models achieved the highest accuracy of almost 99.6%. Additionally, it performed well in terms of training time and prediction execution time, both of which were less than 307 ms [10].

In a similar development, Mohy-Eddine et al. [11], attempted to cut down on computing costs and prediction times, isolation forest and Pearson's correlation coefficient were combined. In order to find and eliminate outliers from datasets, IF was utilized and to determine which features were most suited, the authors used PCC. PCC and IF (PCCIF and IFPCC) were applied interchangeably to improve IDS performance, the RF classifier was used. The Bot-IoT and NF-UNSW-NB15-v2 data were used for evaluation. Notable outcomes are displayed by RF-PCCIF and RF-IFPCC, which have 99.98% and 99.99% Accuracies and 6.18 s and 6.25 s prediction times on Bot-IoT, respectively. Additionally, on the NF-UNSW-NB15-v2 test, the two models scored 99.3% and 99.18% Accuracies and 6.71 s and 6.87 s, respectively, for prediction time. The suggested model has various advantages and performs better than related models, as demonstrated by the results [11]. However, only accuracy

was used to evaluate the model performance, whereas other state-of-the-art metrics that access the robustness of the models needed to be used for the evaluation.

Rather than conducting individual tests for each file on a model-by-model basis, Ref. [12] suggested the entire dataset combined into a single file, subsequently developed a multi-classification labelled dataset by using the attacks families included in the dataset as a new label. Using the improved dataset, the effectiveness of DL models was assessed for two classification categories. The accuracy and loss of the proposed DL model results were compared with those of related research in order to assess the efficacy of DL and ML models in the improved dataset. The accuracy of the suggested DL models produced results in binary classification of 99.26% and multi-class classification of 99.59% [12]. Only accuracy was used to evaluate the model performance, whereas, other state-of-the-art metrics that access the robustness of the models needed to be used for the evaluation.

Reference [13] suggested training and validate the data gathered from TCP/IP packets, the study offered a hybrid rule-based feature selection DL-based IDS for IIoT. The suggested plan was examined using NSL-KDD and UNSW-NB15, two popular network datasets. The results of the performance comparison show that the recommended technique performed better than other related techniques in terms of accuracy, detection rate, and FPR by 99%, 99%, and 1%, respectively, in the NSL-KDD data and 98.9%, 99.9%, and 1.1%, respectively, in the UNSW-NB15 data. Ultimately, simulation studies employing diverse assessment metrics shown that the proposed approach was suitable for IIoT intrusion network attack categorization [13]. Meanwhile, the study requires further investigation for validation as the study conducted mainly on a balanced dataset.

In a different study Tareq et al. [14], two intelligent network methods namely; Inception and DenseNet Time—were trained using a multi-class classification technique to identify cyberattacks. The ToN-IoT dataset, which contains heterogeneous data; the UNSW2015 and Edge-IIoT datasets were the three datasets used by the study to measure the performance of these two networks. After that, a number of cyberattacks were identified and the outcomes were compared. On standard ToN-IoT datasets, extensive experiments were carried out with the DenseNet Multcategorization method. The Inception Time method produced the best result for Windows 10 with the network, with 100% accuracy, while the DenseNet approach yielded the best accuracy of 99.9%. The best outcome when using the Inception Time method to the Edge-IIoT dataset was an accuracy of 94.94%. The Inception Time technique yielded a 98.4% accuracy rate when used to evaluate the attacks in the UNSW-NB15 database. A marginal improvement was observed when the Inception Time model was trained using window sequences for the sliding window technique and a six-window size to begin with. The model's multi-category classification accuracy rate was 98.6% [14]. In the main time, the study requires further investigation for validation as the study conducted mainly on a balanced dataset.

Bushra et al. [15] suggested a hybrid seagull optimized ResNet 101-C (HSO-ResNet 101-C) technique was developed to accurately detect different kinds of attacks. With the use of the HSO-ResNet 101-C architecture's computational complexity—which demands more time and resources while handling large data—was resolved. Accuracy, recall, precision, and F1-score are a few of the performance indicators used to enhance ID performance in big data contexts. The IDs rate may be found using the two types of datasets, CICIDS2017 and UNSW-NB15. For the CICIDS2017 dataset, the accuracy, precision, recall, and F1-score FAR parameters yielded performance rates of 100%, 99.88%, 99.2%, 98.9%, and 6.38%, respectively. The UNSW-NB15 dataset yielded performance rates of 98.9%, 95%, 96.4%, and 97.8% for accuracy, recall, precision, and F1-score FAR respectively. The success of this study may relay on the optimization method used.

In Ref. [16], the UNSW-NB15 dataset was utilized in 2021 to train machine learning classifiers. Various classifiers, including KNN, stochastic gradient descent (SGD), RF, LR, and Naïve Bayes (NB) classifiers, were trained using the taxonomy of classifiers namely designed for lazy and eager learners. Using Chi-Square, a filter-based feature selection method, the UNSW-NB15 dataset was analyzed to eliminate unnecessary and redundant characteristics. The focus of the study was on feature selection.

Reference [17] aimed to compare the use of several Machine Learning algorithms in constructing and enhancing intrusion detection systems. The evaluation was based on several metrics including confusion matrix, accuracy, recall, precision, FAR, specificity, and sensitivity. The UNSW-NB15 Dataset was used for this purpose. No experimental results were reported by the authors.

Reference [18] used the UNSW-NB 15 dataset as an offline dataset to assess the performance of machine learning models based on binary classification. The effectiveness of the DT, RF, gradient boosting tree (GBT), and MLP were evaluated through testing. The study employed the Chi-Square test to eliminate factors that were not independently related to the response. It was determined that DT is the most effective classifier with the highest accuracy and the lowest false positive rate (FPR). All models, except for RF, showed enhanced overall performance after feature reduction. Empirical investigation demonstrated that our suggested methodology outperformed other established machine learning techniques in terms of precision. However, the study was limited to only four ML models while the current study analysis six ML models.

Methodology

The necessary procedures and methodological stages to complete this research have been illustrated in this section. The research methodology is chronologically explained in this section, this includes research design, research procedure how data was acquired, preprocessed, modelling and tested which is similar to [19–21]. Figure 1 presents the framework of comparative analysis of ML models for IDS.

Acquired Dataset and Description

The UNSW-NB15 data sets' raw network packets are configured at the cyber range lab at the Austrian Centre for Cyber Security. The same dataset was resourceful in [22, 23] among other studies in the area of IDS and security at large. This configuration is a combination of real-time routine operations and simulated contemporary attack behavior utilizing the IXIA Perfect-storm technique. The tcpdump software can be used to save 100 GB of unprocessed network data. To generate a total of 49 features along with the class label, twelve algorithms are utilized, and the Argus and Bro-IDS tools are deployed. The original dataset consists of four CSV files, which collectively include a total of 2,540,044 packets [24]. The majority of academics have employed these data to evaluate each element of their developed IDS separately. The UNSW Dataset was downloaded from a public repository called Kaggle. The dataset was a unbalanced dataset since the difference between the classes (normal and attack) is as much as 84%. In other words, the dataset has 92.71% normal and 7.29% attack.

Data Pre-processing

Mainly in this study, data pre-processing includes data cleaning, data normalization and data splitting as described thus:

Data Cleaning

Since the original dataset cannot be used to feed the ML models because it has a large number of missing values in several feature columns. During the data cleaning stage, the following features were removed from the dataset. The IDS sample dataset used for this study is found in Table 1.

Data Normalization

Data normalization was particularly useful for systems where measurements are frequently reported on widely disparate

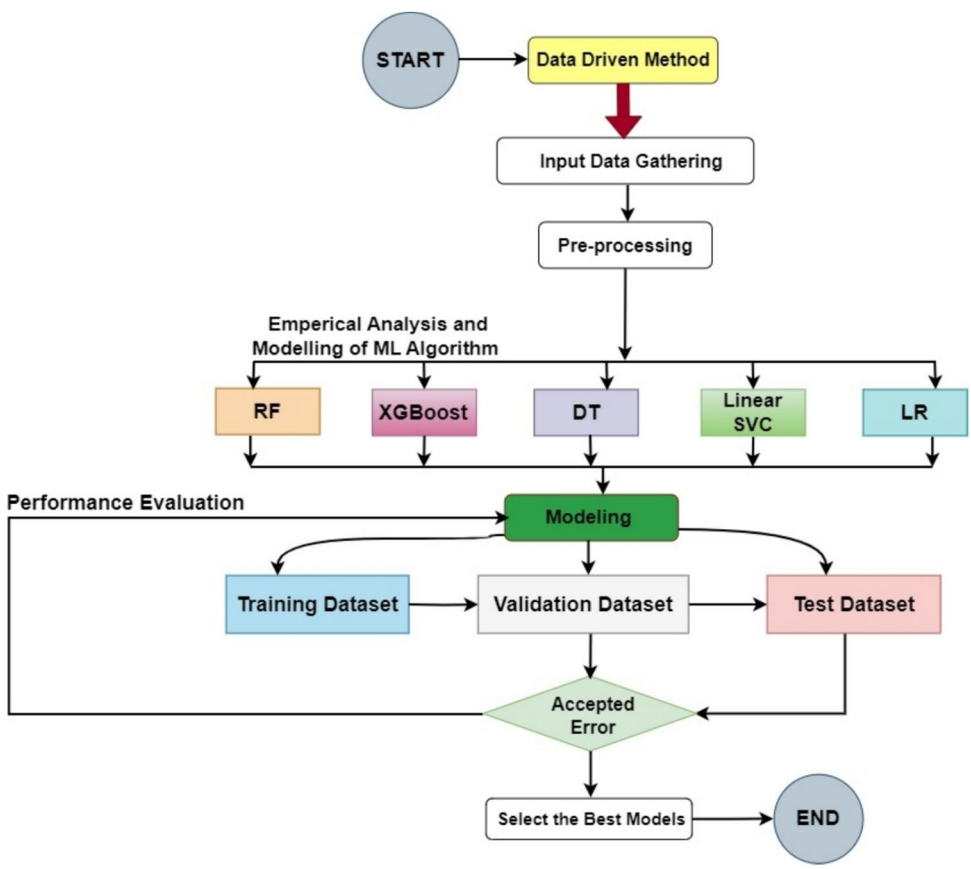

Fig. 1 Framework for ML-based intrusion detection comparative study

Table 1 IDS sample dataset

'StartTime', 'LastTime', 'SrcAddr', 'DstAddr', 'sIpId', 'dIpId', as they are unique to the attacks and would expose the type of the attack to the model; therefore, the model would not be generalized for unseen data. The dependent variable contains multiple classes which are normal, DoS, Reconn, Backbone and CommInj and later converted to binary classification problem (normal and attack). Also, the categorical variables in the dataset were converted to numerical with aid of python library called LabelEncoder.

levels. More reliably constructed neural networks are made possible by min–max normalization. The advantage of this normalization technique is that it precisely preserves all data linkages, meaning it does not lead to bias. While min–max is added, the rising function remains below the proper value range for the classification, but the corresponding distributions of the associated features remain inside the current value range [25].

Data Splitting

At this point, the primary dataset was divided into three groups: a test set (15%), a validation set (15%), and a training set (70%).

Modelling

We suggested comparing five different types of modern ML models for an IDS viz-a-viz linear SVC, LR, DT, RF, and XGB in this study. Therefore, the dataset was subjected to random sampling using an oversampling technique in order to improve its compatibility with the machine learning model.

Performance Evaluation Measures

This will assess and gauge the effectiveness of ML models for IDS. Additionally, the writers evaluated each model's performance using accuracy and loss criteria. Additionally, because a callback function is used to track validation loss at each training epoch, the training process will end if validation loss has not improved after twenty epochs. After the learning process is complete, the test sets were used to assess the ML model. Assessing the ML models based on their ability to accurately identify, recall, and provide precise results, as well as their F1-score and confusion matrix. The mathematical formulations for the performance metrics utilized in this investigation are shown in Eqs. 1–4, as specified by reference [26].

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{1}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

$$\text{Recall (TP Rate)} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

$$\text{F1} - \text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

Result and Discussion

This section presents and explains the results of the research and at the same time give the comprehensive discussion. The results were illustrated in different figures as presented [14] for a better understating of the models' performances. The findings of the study are discussed, and the results are presented in this part. Using a Jupyter notebook and sklearn libraries on the Anaconda platform in python, the experiment was carried out. An Intel® CoreTM i5-7200 CPU running at 2.50–2.70 GHz on a Pentium Windows PC with 8-GB RAM was used for all trials. To get rid of certain undesired background pictures and noise, the photographs were manually cropped.

Experimental Results

As a result of doing the Exploratory Data Analysis (EDA) on the UNSW-NB15 dataset, the results of the experiment reveal that the proportion of normal traffic is 92.71% (0, normal), whereas the percentage of attack traffic is 7.29% (1, attack). This is illustrated in Fig. 2 and it shows the unbalance nature of the intrusion dataset in this study. In the process of oversampling, synthetic data is generated in order to randomly choose attributes from observations belonging to the minority class. On the other hand, undersampling entails randomly deleting some observations from the majority class in order to achieve a balance between the numbers belonging to the minority class and the majority class. It was discussed in Sect. 4.2 that the results of a comparison of the performances of four different machine learning models on the dataset that had been extracted and reduced were shown.

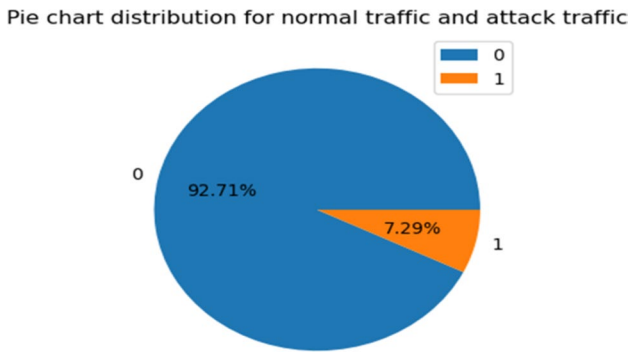

Fig. 2 The EDA on the UNSW-NB15 dataset for the ML comparative study

All of the metrics have a value that falls somewhere between '0' and '1'. The more closely the values of the measures are located to 1, the better the model is.

The results of the modern machine learning models that were applied to the dataset are depicted in Figs. 3 and 4, respectively. The results of an evaluation of the performance of several classifiers over the UNSW dataset are presented in Fig. 3. The performance of five different classifiers, namely Lin-ear SVC, LR, RF, DT, and XGBoost, is evaluated and compared using four different measures, including Accuracy, Recall, Precision, and F1- Score. To determine whether or not the model is accurate in general, the accuracy metrics are used. The accuracy of all classifiers is relatively high, with RF and XGBoost demonstrating slightly better performance than the others. In a similar fashion, recall metrics are used to evaluate the capability of the technique to accurately identify all instances that should be considered significant. When it comes to identifying real positive cases, RF and XGBoost have the highest recall, which indicates that they are superior to other methods. The capacity of the method to appropriately identify just occurrences that are relevant is what we mean when we talk about precision. The RF and XGBoost algorithms, once again, have the highest precision, which indicates that they produce fewer false positives. As a last point of discussion, the F1-score is the harmonic mean of precision and recall, which offers a single metric for evaluating the performance of the model. By displaying

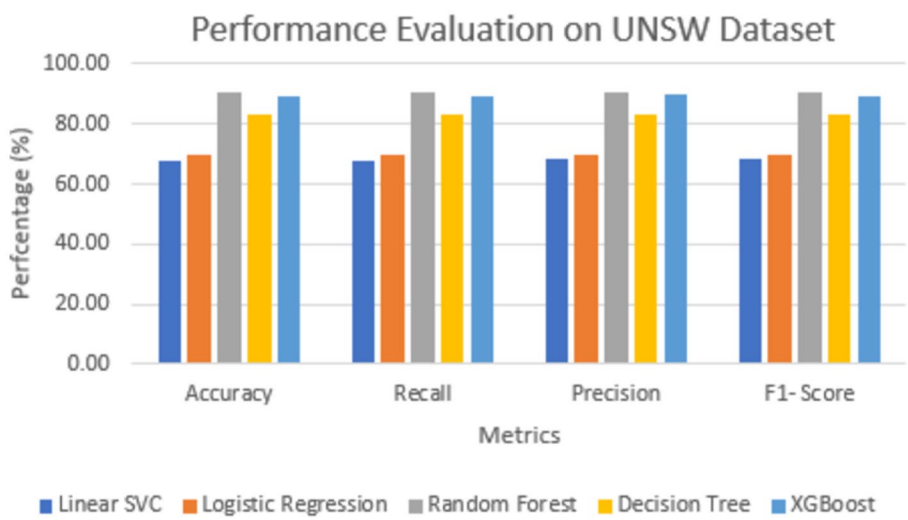

Fig. 3 Result of the contemporary ML models in IDS experimental study

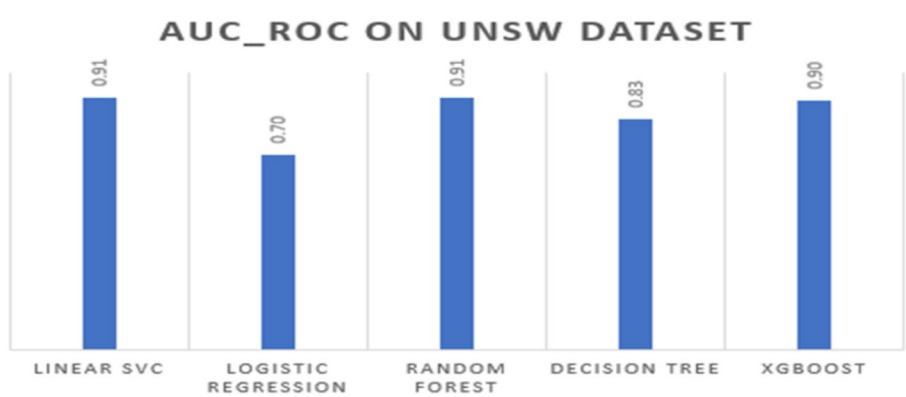

Fig. 4 AUC_ROC scores on UNWS dataset for IDS study using ML models

the greatest F1-scores, RF and XGBoost demonstrate that they have achieved a balance between precision and recall.

The AUC_ROC for the same group of classifiers is shown in Fig. 4, which is based on the UNSW Dataset. With an area under the receiver operating characteristic curve (AUC-ROC) score of 0.91, Linear SVC and RF have the highest performance in differentiating between the classes. With an area under the receiver operating characteristic curve (AUC-ROC) score of 0.90, XGBoost demonstrates a high level of discriminatory power. The AUC-ROC score for decision tree is 0.83, which is satisfactory but lower than the scores for Linear SVC, RF, and XGBoost techniques. With an area under the receiver operating characteristic curve (AUC-ROC) score of 0.70, logistic regression is the classifier that has the lowest discriminatory power among those that were tested. The classifiers RF and XGBoost perform exceptionally well across all measures, with excellent accuracy, recall, precision, F1-score, and area under the receiver operating characteristic curve (AUC-ROC). Because of this, they are highly reliable options for the UNSW dataset. In addition, the Linear SVC performs admirably, particularly in terms of AUC-ROC; but, its precision and F1-score are marginally inferior to those of the RF and XGBoost algorithms. The performance of decision tree is satisfactory, although it is not as impressive as the performance of the top performers in this study. Among the several classifiers, logistic regression has the least effective performance, notably in terms of AUC-ROC, which indicates that it is less effective for this dataset than the other candidates. According to these findings, ensemble approaches like as RF and XGBoost deliver the greatest performance for the UNSW dataset. As a result, these methods are ideal alternatives for tasks that need high classification accuracy and robustness.

The confusion matrices for the five machine learning models that were tested in this investigation are displayed in Figs. 5, 6, 7, 8, 9. The figures illustrate how well the machine learning models performed on a set of test data for which the actual values were known. Those specialists who would be interested in utilizing these models for an IDS deployment will find this to be helpful. It makes the visualization of the performance of the model easier to accomplish. With the use of count values, a summary of the numbers of cases that were correctly and wrongly classified may be obtained. These numbers are then broken down according to class (attack and normal). In addition to shedding information on the errors that are being made by the model, the primary objective of the confusion matrix performance evaluation is to shed light on the types of errors that are being made. The ROC curve was utilized in order to ascertain the degree to which the models were able to differentiate between normal and attack state. If the value is close to 1, the ROC value falls between '0' and '1', which suggests that the model's

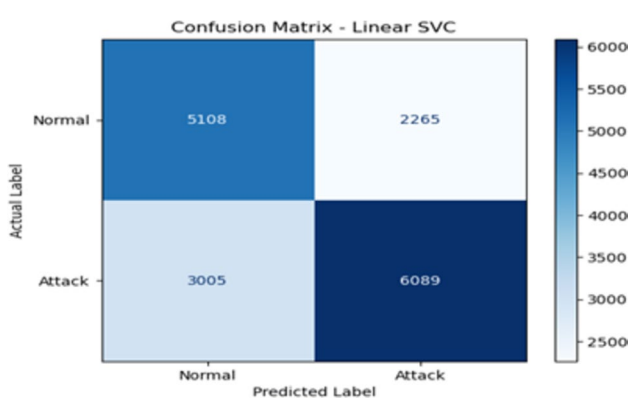

Fig. 5 Confusion matrix for Linear SVC

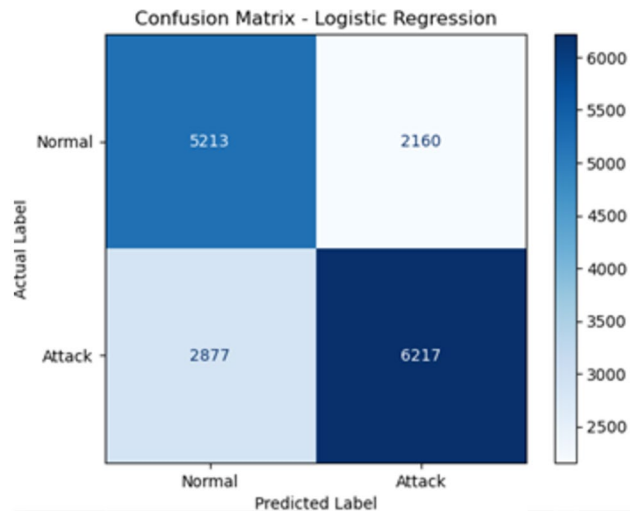

Fig. 6 Confusion matrix for LR

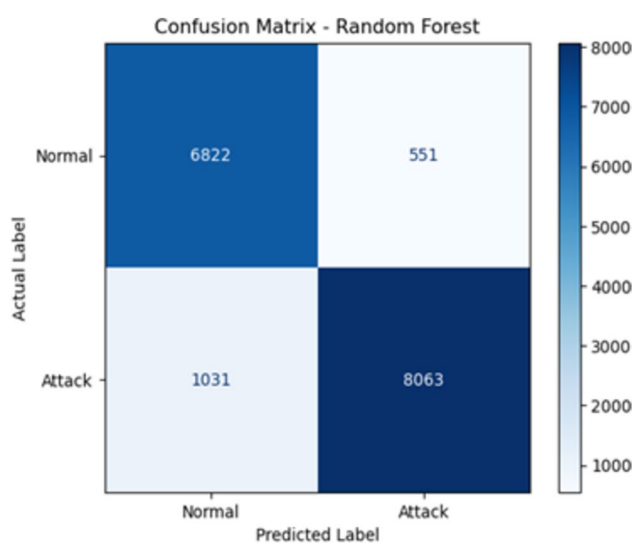

Fig. 7 Confusion matrix for RF

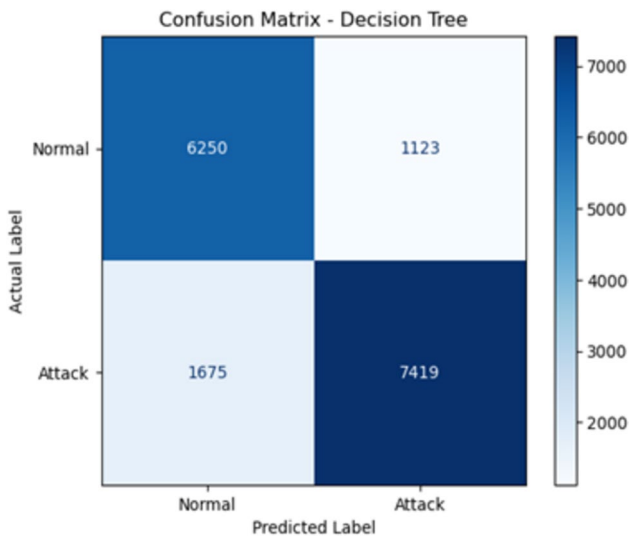

Fig. 8 Confusion matrix for DT

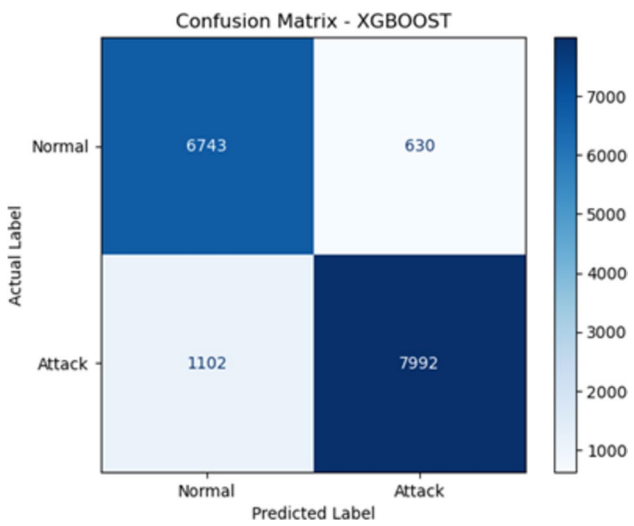

Fig. 9 Confusion matrix for XGB

measure of separability is good. On the other hand, if the value is close to 0, it indicates that the model is the worst possible measure of separability.

Discussion

Furthermore, this study includes experimental data that were previously mentioned in the preceding section. The accuracy of Linear SVC, LR, RF, DT, and XGB was assessed and the results for the train set, and test set are shown in Tables 2 and 3, respectively. The dataset was a unbalanced dataset since the difference between the classes (normal and attack) is as much as 84% (92% to 8%) when compared with the UNSW dataset. The performance evaluation shows that RF has high performance followed by the XGBoost model. Figures 5, 6, 7, 8, 9a, b represent the performance metrics of Linear SVC, LR, RF, DT and XGBoost classifier used for a binary classification task. The overall results suggest that the XGBoost classifier is effective for the given classification task, providing reliable performance in distinguishing between normal and attack instances.

Table 2 Train Set results of ML models in IDS task

ML models	Accuracy	Recall	Precision	F1-score
Linear SVC	68.72	68.72	69.21	68.80
Logistic regression	69.92	69.92	70.36	69.99
Random forest	97.87	97.87	97.93	97.87
Decision tree	83.68	83.68	83.96	83.37
XGBoost	92.14	92.14	92.30	92.16

Table 3 Test set results of ML models in IDS task

ML models	Accuracy	Recall	Precision	F1-score	AUC ROC
Linear SVC	68.00	68.00	68.44	68.08	0.91
Logistic regression	69.41	69.81	69.84	69.49	0.70
Random forest	90.39	90.39	90.58	90.41	0.91
Decision tree	83.01	83.00	83.28	83.05	0.83
XGBoost	89.48	89.48	89.68	89.51	0.90

The XGBoost classifier shows better output with an AUC of 0.90, indicating a high ability to discriminate between the classes. The confusion matrix provides additional evidence by displaying a substantial count of accurate positive and negative predictions, with relatively fewer false positives and false negatives. The calculated metrics are all high, particularly the precision, which means that when the classifier predicts an attack, it is correct 93% of the time. The recall of 88% indicates that it successfully identifies 88% of actual attacks. The ROC curve has an area of 0.90, which signifies a strong level of classification performance. AUC, or area under the curve, is a metric used to evaluate the performance of a model. A value of 1 indicates a flawless model, while a value of 0.5 indicates a model that cannot distinguish between classes. Thus, an AUC value of 0.90 indicates that the classifier possesses a remarkable capacity to differentiate between the two groups.

Comparison with the Existing Studies and the Current Study

In Table 4, we compare the existing studies and current studies to draw the contributions and importance of this study. Therefore, ML models considerations of related studies, models training and testing performances provided in each

Table 4 Comparison of existing studies and the current study

References	ML consideration for IDS	Models training performance	Models testing performance
Reference [8]	k-NN, SVM, NB, RF, DT, and SGD	No	Yes
Reference [10]	KNN, SVM, RF, DT, Gaussian NB, XGB, LR, artificial neural networks, and MLP	No	Yes
Reference [18]	DT, RF, GBT, and MLP	No	Yes
Reference [16]	KNN, SGD, RF, LR, and NB classifiers	No	Yes
Current study	Linear SVC, LR, RF, DT and XGBoost	Yes	Yes

of these studies were presented. It was observed that existing studies have not been proving models training performances, which is equally important to make decisions and the computational complexity of ML models during training varies from testing.

Conclusion

Network IDS are widely regarded as a very efficient and effective method for addressing network security threats. In addition, machine learning models in established IDS are utilized through a highly effective and efficient detection process to address the growing complexity of cyber security threats. The implementation of a precise and effective system that detects and mitigates malicious attacks is now a must in order to minimize the security risks connected with IoT devices. Several scholars have utilized a variety of methodologies, ranging from basic codes to algorithms and models. This research work has performed comparative experiments using various ML models such as Linear SVC, LR, RF, DT and XGB on the UNSW NB15 datasets to detect intrusion. Performance evaluation results showed that RF and XGBoost outperformed the other ML models. Hence, ML models can effectively be used to detect system attacks. This research may be expanded in the future and use the paradigm in a real-world dataset with further conclusions and justifications. In addition, the current research provides both indebt analysis of the training and testing performances of the ML models under consideration. However, deep learning models may be used to gain better insight into the dataset, and deep learning may be compared with machine learning in the future research.

Acknowledgements Authors acknowledge the Centre of Excellence, University of Zululand for the support received for this work.

Funding Open access funding provided by University of Zululand.

Data availability The data availability statement is not applicable.

Declarations

Conflict of interest Authors have no competing interest to declare on this work.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Vitorino J, Andrade R, Praça I, Sousa O, Maia E. A Comparative analysis of machine learning techniques for IoT intrusion detection. In: Aïmeur E, Laurent M, Yaich R, Dupont B, Garcia-Alfaro J, editors, Foundations and practice of security. FPS 2021, Lecture notes in computer science, vol 13291. Cham: Springer; 2022. https://doi.org/10.1007/978-3-031-08147-7_13. p. 191–207.
2. Sisinni E, Saifullah A, Han S, Jennehag U, Gidlund M. Industrial internet of things: challenges, opportunities, and directions. IEEE Trans Ind Inform. 2018;14(11):4724–34. https://doi.org/10.1109/TII.2018.2852491.
3. Benamor Z, Seghir ZA, Djezzar M, Hemam M. A comparative study of machine learning algorithms for intrusion detection in IoT network. Revue d'Intelligence Artificielle. 2023;37(3):567–76.
4. Ajagbe SA, Florez H, Awotunde JB. AESRSA: a new cryptography key for electronic health record security. In: Florez H, Gomez H, editors. Communications in computer and information science, vol 1643. Peru: Springer; 2022. p. 244–58.
5. Gautam RK, Doegar EA. An ensemble approach for intrusion detection system using machine learning algorithms. In: International conference on cloud computing, data science & engineering (confluence). 2018. p. 14–5.
6. Tahri R, Jarrar A, Lasbahani A, Balouki Y. A comparative study of machine learning algorithms on the UNSW-NB 15 dataset. ITM Web Conf. 2022;48:03022. https://doi.org/10.1051/itmconf/20224803002.
7. Ajagbe SA, Adegun AA, Mudali P, Adigun MO. Performance of machine learning models for pandemic detection using COVID-19 dataset. In: 2023 IEEE AFRICON. Kenya: IEEE; 2023. https://doi.org/10.1109/AFRICON55910.2023.10293525. p. 1–6.

8. Saran N, Kesswani N. A comparative study of supervised machine learning classifiers for intrusion detection in internet of things. Proc Comput Sci. 2023;218:2049–57.
9. Nizamudeen SM. Intelligent intrusion detection framework for multi-clouds—IoT environment using swarm-based deep learning classifier. J Cloud Comput. 2023;12:134. https://doi.org/10.1186/s13677-023-00509-4.
10. Rbah Y, Mahfoudi M, Balboul Y, Chetioui K, Fattah M, Mazer S, Bernoussi B. A machine learning based intrusions detection for IoT botnet attacks. AIP Conf Proc. 2023;2814: 030012. https://doi.org/10.1063/5.0149102.
11. Mohy-Eddine M, Guezzaz A, Benkirane S, Azrour M, Farhaoui Y. An ensemble learning based intrusion detection model for industrial IoT security. Big Data Min Anal. 2023;6(3):273–87. https://doi.org/10.26599/BDMA.2022.9020032.
12. Aleesa AM, Younis MI, Mohammed AA, Sahar NM. Deep-intrusion detection system with enhanced Unsw-Nb15 dataset based on deep learning techniques. J Eng Sci Technol. 2021;16(1):711–27.
13. Awotunde JB, Chakraborty C, Adeniyi AE. Intrusion detection in industrial internet of things network-based on deep learning model with rule-based feature selection. Wirel Commun Mob Comput. 2021;2021:1–17. https://doi.org/10.1155/2021/7154587.
14. Tareq I, Elbagoury BM, El-Regaily S, El-Horbaty E-S. Analysis of ToN-IoT, UNW-NB15, and edge-IIoT datasets using DL in cybersecurity for IoT. Appl Sci. 2022;12(19):9572. https://doi.org/10.3390/app12199572.
15. Bushra SN, Subramanian N, Chandrasekar A. An optimal and secure environment for intrusion detection using hybrid optimization based ResNet 101-C model. Peer-to-Peer Netw Appl. 2023;16:2307–24. https://doi.org/10.1007/s12083-023-01500-1.
16. More S, Idrissi M, Mahmoud H, Asyhari AT. Enhanced intrusion detection systems performance with UNSW-NB15 data analysis. Algorithms. 2024;17(2):64. https://doi.org/10.3390/a17020064.
17. Rosy JV, Kumar SB. Intrusion detection on the Unsw-Nb15 dataset using feature. Webology. 2021;18(6):4784–802.
18. Shushlevska M, Efnusheva D, Jakimovski G, Todorov Z. Anomaly detection with various machine learning classification techniques over UNSW-NB15 Dataset. In: Proc. of the 10th international conference on applied innovations in IT, (ICAIIT). 2022. p. 21–7.
19. Ajagbe SA, Adegun AA, Olanrewaju AB, Oladosu JB, Adigun MO. Performance investigation of two-stage detection techniques using traffic light detection dataset. IAES Int J Artif Intell (IJ-AI). 2023;12(4):1909–19. https://doi.org/10.11591/ijai.v12.i4.pp1909-1919.
20. Olanloye O, Olasunkanmİ O, Oduntan OE. Comparison of support vector machine models in the classification of susceptibility to schistosomiasis. Balkan J Electr Comput Eng. 2020;8(3):266–71.
21. Moustafa N, Slay J. A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set). In: 2015 Military communications and information systems conference (MilCIS), Canberra. 2015. p. 1–6.
22. Dickson A, Thomas C. Analysis of UNSW-NB15 dataset using machine learning classifiers. In: Thampi SM, Piramuthu S, Li KC, Berretti S, Wozniak M, Singh D, editors. Machine learning and metaheuristics algorithms, and applications, communications in computer and information science, vol 1366. https://doi.org/10.1007/978-981-16-0419-5_16. Singapore: Springer; 2021. p. SoMMA 2020.
23. Disha RA, Waheed S. A comparative study of machine learning models for network intrusion detection system using UNSW-NB 15 dataset. In: 2021 international conference on electronics, communications and information technology (ICECIT). Khulna: IEEE. 2021. p. 1–5. https://doi.org/10.1109/ICECIT54077.2021.9641471.
24. Myint H, Meesad P. Incremental learning algorithm based on support vector machine with Mahalanobis distance (ISVMM) for intrusion prevention. In: 6th international conference on electrical engineering/electronics, computer, telecommunications and information technology, Pattaya. 2009. p. 630–33.
25. Gaber T, Awotunde JB, Folorunso SO, Ajagbe SA, Eldesouky E. IIoT intrusion detection method based on machine learning and optimization techniques. Wirel Commun Mob Comput. 2023;2023:1–15. https://doi.org/10.1155/2023/3939895.
26. Jooda JO, Ogundokun RO, Adedeji TO, Ajagbe SA, Adigun MO, Oke AO, O OE. Multiple instances fingerprint image data acquisition. In: 2023 international conference on electrical, computer and energy technologies (ICECET). Cape Town: IEEE; 2023. p. 1–6. https://doi.org/10.1109/ICECET58911.2023.10389447.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.